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CRUD OPERATIONS:(MYSQL)

```
package CRUD;
```

```
import java.sql.Connection;
```

```
import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;
```

```
public class CRUD_Test {
```

```
    public void create_data(String sl_no,String name,String email){
```

```
        DB_Connection obj_DB_Connection=new DB_Connection();
```

```
        Connection connection=obj_DB_Connection.get_connection();
```

```
        PreparedStatement ps=null;
```

```
        try {
```

```
            String query="insert into student(sl_no,name,email) values (?,?,?)";;
```

```
            ps=connection.prepareStatement(query);
```

```
            ps.setString(1,sl_no);
```

```
            ps.setString(2,name);
```

```
            ps.setString(3,email);
```

```
            System.out.println(ps);
```

```
            ps.executeUpdate();
```

```
        } catch (Exception e) {
```

```
            System.out.println(e);
```

```
        }
```

```
    }
```

```
    public void read_data(String sl_no) {
```

```
        DB_Connection obj_DB_Connection = new DB_Connection();
```

```
        Connection connection = obj_DB_Connection.get_connection();
```

```
        PreparedStatement ps = null;
```

```
        ResultSet rs = null;
```

```
        try {
```

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```
String query = "SELECT * FROM student WHERE sl_no =  
?"; ps = connection.prepareStatement(query);  
  
ps.setString(1, sl_no);  
  
System.out.println(ps);  
  
rs = ps.executeQuery();  
  
if (rs.next()) {  
    System.out.println("Sl no - " + rs.getString("sl_no"));  
    System.out.println("Name - " + rs.getString("name"));  
    System.out.println("Email - " + rs.getString("email"));  
} else {  
    System.out.println("No record found for sl_no: " + sl_no);  
}  
  
} catch (Exception e) {  
    System.out.println(e);  
}  
}  
  
//  
public void update_data(String sl_no,String new_sl_no,String name,String email){  
    DB_Connection obj_DB_Connection=new DB_Connection();  
    Connection connection=obj_DB_Connection.get_connection();  
    PreparedStatement ps=null;  
    try {  
        String query="update student set sl_no=?,name=?,email=? where  
sl_no=?";
```

```
        ps=connection.prepareStatement(query);

        ps.setString(1, new_sl_no);

        ps.setString(2, name);

        ps.setString(3, email);

        ps.setString(4, sl_no);

        System.out.println(ps);

        ps.executeUpdate();

    } catch (Exception e) {

        System.out.println(e);

    }

}

//

public void delete_data(String sl_no){

    DB_Connection obj_DB_Connection=new DB_Connection();

    Connection connection=obj_DB_Connection.get_connection();

    PreparedStatement ps=null;

    try {

        String query="delete from student where sl_no=?";

        ps=connection.prepareStatement(query); ps.setString(1,

        sl_no);

        System.out.println(ps);

        ps.executeUpdate();

    } catch (Exception e) {

        System.out.println(e);

    }

}

public void alter_table() {

    DB_Connection obj_DB_Connection = new DB_Connection();

    Connection connection = obj_DB_Connection.get_connection();
```

```
        PreparedStatement ps = null;

        try {

            String query = "alter table student ADD City
            VARCHAR(20)"; ps =
            connection.prepareStatement(query);

            System.out.println(ps);

            ps.executeUpdate();
            System.out.println("Column added successfully.");

        } catch (Exception e) {
            System.out.println(e);
        }
    }

    public void truncate_data() {
        DB_Connection obj_DB_Connection = new DB_Connection();
        Connection connection = obj_DB_Connection.get_connection();
        PreparedStatement ps = null;

        try {
            String query = "truncate table student";
            ps = connection.prepareStatement(query);

            System.out.println(ps);

            ps.executeUpdate();
            System.out.println("All records deleted successfully.");
```

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```
        } catch (Exception e) {

            System.out.println(e);

        }

    }

    public static void main(String[] args) {

        CRUD_Test objTest=new CRUD_Test();

        //insert data

        objTest.create_data("1", "jinu", "jinu@gmail.com");

        objTest.create_data("2", "jawad", "jawad@gmail.com");

        // read data

        objTest.read_data("2");

        //update data

        objTest.update_data("2", "22", "jawad2", "jawad2@gmail");

        //delete data

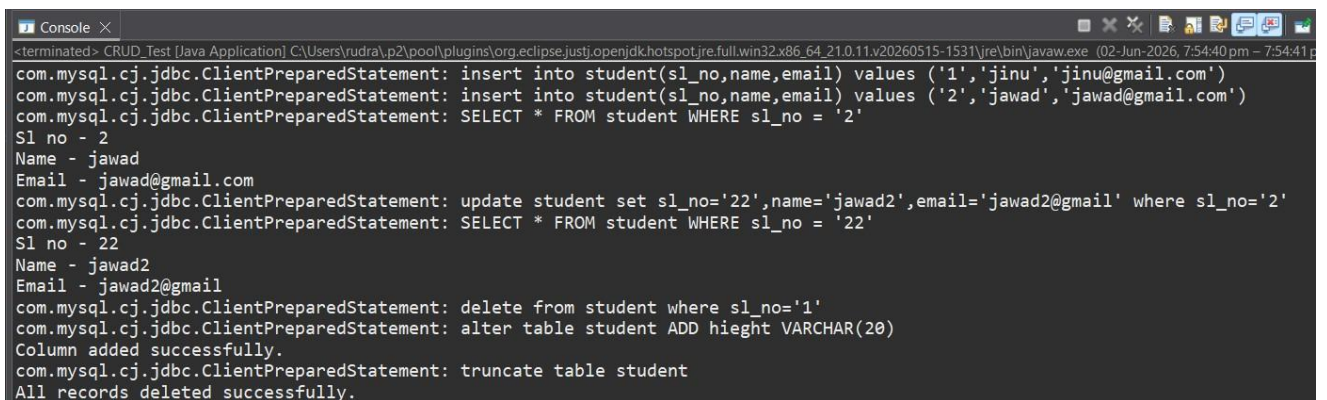
        objTest.delete_data("1");

        objTest.alter_table();

        objTest.truncate_data();

    }

}
```



```
Console X
<terminated> CRUD_Test [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:54:40 pm - 7:54:41 p
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(s1_no,name,email) values ('1','jinu','jinu@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(s1_no,name,email) values ('2','jawad','jawad@gmail.com')
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE s1_no = '2'
S1 no - 2
Name - jawad
Email - jawad@gmail.com
com.mysql.cj.jdbc.ClientPreparedStatement: update student set s1_no='22',name='jawad2',email='jawad2@gmail' where s1_no='2'
com.mysql.cj.jdbc.ClientPreparedStatement: SELECT * FROM student WHERE s1_no = '22'
S1 no - 22
Name - jawad2
Email - jawad2@gmail
com.mysql.cj.jdbc.ClientPreparedStatement: delete from student where s1_no='1'
com.mysql.cj.jdbc.ClientPreparedStatement: alter table student ADD hieght VARCHAR(20)
Column added successfully.
com.mysql.cj.jdbc.ClientPreparedStatement: truncate table student
All records deleted successfully.
```

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MongoDB Connection:

```
package mongodb2;

import com.mongodb.MongoClient;

import com.mongodb.client.MongoDatabase;

public class Test1 {

    public static void main(String[] args) {

        MongoClient mongoClient =

            new MongoClient("localhost", 27017);

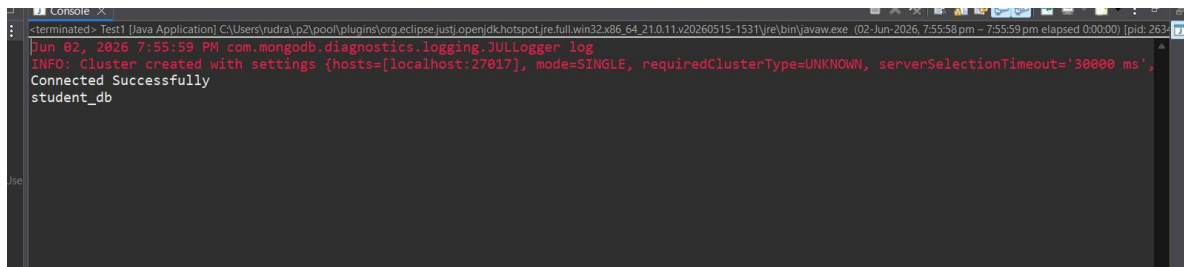
        MongoDatabase database =

            mongoClient.getDatabase("student_db"); System.out.println("Connected

            Successfully"); System.out.println(database.getName());

    }

}
```



```
<terminated> Test1 [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk hotspot.jre.full.win32.x86_64.21.0.11.v20260515-1531\jre\bin\javaw.exe (02-Jun-2026, 7:55:58 pm - 7:55:59 pm elapsed 0:00:00) [pid: 263]
Jun 02, 2026 7:55:59 PM com.mongodb.diagnostics.logging.JULLogger log
INFO: Cluster created with settings {hosts=[localhost:27017], mode=SINGLE, requiredClusterType=UNKNOWN, serverSelectionTimeout='30000 ms',
Connected Successfully
student_db
```